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G. Physics-I Mid-Exam

1. Give short answers for the following questions (9pts)

1. The particle is launched vertically upward with its initial velocity v_0 from the ground, the time required for the particle to hit the ground is $\frac{2v_0}{g}$ or $\frac{x}{v_0}$ (2pts)

2. If a car is traveling westward with a constant speed of 20 m/s, The resultant force acting on it is 0N (1pts)

3. If the speed of a particle is doubled, by what factor is its momentum changed? By what factor is its kinetic energy changed? (2pts)

given $v_f = 2v_i$ factor of momentum = $\frac{mv_f}{mv_i} = \frac{m(2v_i)}{mv_i} = 2$ where $v_i = 20$
 proportionality $\frac{mv_f}{mv_i} = \frac{m(2v_i)}{mv_i} = 2$
 change of momentum = 2
 by factor of 2 (times)
 kinetic energy $k = \frac{1}{2}mv^2$
 $k_i = \frac{1}{2}m(v_i)^2$
 $k_f = \frac{1}{2}m(2v_i)^2 = \frac{1}{2}m(4v_i^2) = 4 \times \frac{1}{2}mv_i^2 = 4k_i$
 kinetic energy changed by factor of 4 (times)

4. State and Explain? (Each 1pts)

a. work energy theorems

$E_{work} = F \cdot d$ - there are many types of work such as gravitational work done against gravitational force
 work due to kinetic
 work done to decrease friction
 work done due to spring

Energy is neither created nor destroyed and cannot be destroyed
 there is also maximum work and minimum work done depending on given condition

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b. Newton's Laws of motion

Newton first law
Newton second law
Newton third law

① Newton first law state that the particle at rest remain at rest unless an unbalanced force acting on it and the particle in motion continues in its motion unless an unbalanced force acting on it.

② Newton second law state that $\vec{F} = m\vec{a}$ where Force $\vec{F}$ is in the direction of acceleration, $\vec{a}$. Force, F is directly proportional to acceleration a and is inversely proportional to mass, m .

Newton third law state that action and reaction force which are equal and opposite and their direction is both force acts on different body in the same line.

c. The properties of conservative force.

Work done by conservative force does not depend on path taken but it depends on initial and final position. Under conservative force mechanical energy is conserved. Examples of conservative forces are force due to gravity, force due to spring and the electric.

d. Center of mass of the system of particles

Center of mass of particles is as its name explain the balancing point of all masses at a given point and also we can say center of mass is the location of mass at given instant.

11. workout the following problems. Show the necessary steps (11pts)

1. The position of an electron is given $r = 3ti - 4t^2j + 2k$ (3pts)

- a) What is the displacement of the electron at $t = 1s$ to $t = 4s$ 3 - 52
3 - 81
 b) What are the magnitude and direction of velocity of electron $t = 3s$ 2 - 24
1 - 16
3i
 b) What is the average acceleration between $t = 1s$ to $t = 3s$
 c) The instantaneous acceleration at any time t

given in part (a)
 $r(t) = 3ti - 4t^2j + 2k$
 $t_1 = 1s$
 $t_2 = 4s$
 $\Delta r = r_2 - r_1$ (at $t_1 = 1s$)
 $r_1 = 3(1)i - 4(1)^2j + 2k = 3i - 4j + 2k$
 $r_2 = 3(4)i - 4(4)^2j + 2k = 12i - 64j + 2k$
 $\Delta r = (12i - 64j + 2k) - (3i - 4j + 2k) = 9i - 60j$

$V_{at t=4s} = 4s$
 $V = 3(4)i - 4(4)^2j + 2k = 12i - 64j + 2k$
 $\Delta V = V_2 - V_1 = (12i - 64j + 2k) - (3i - 4j + 2k) = 9i - 60j$
 $\Delta t = 3s$
 $a = \frac{\Delta V}{\Delta t} = \frac{9i - 60j}{3} = 3i - 20j$

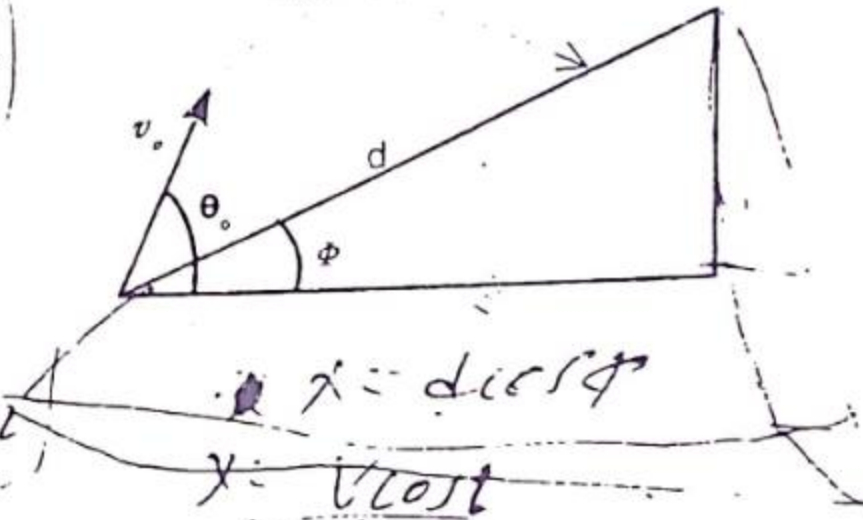
given in part (b)
 $r = 3ti - 4t^2j + 2k$
 $t = 3s$
 $V = \frac{dr}{dt} = 3i - 8tj$
 $V = 3i - 8(3)j = 3i - 24j$
 $|V| = \sqrt{3^2 + 24^2} = \sqrt{585} = 24.19m/s$
 $\theta = \tan^{-1}(\frac{-24}{3}) = -89.1^\circ$

$V = 3i - 8tj$
 $V = 3i - 24j$
 $|V| = \sqrt{3^2 + 24^2} = \sqrt{585} = 24.19m/s$
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2. A projectile is fired up (incline angle ϕ) with an initial speed v_0 at an angle θ_0 with respect to the horizontal ($\theta_0 > \phi$) as shown in Figure below. Show that the projectile travels a distance d up the incline, where (3pts)

$$d = \left(\frac{2v_0^2 \cos \theta_0 \sin(\theta_0 - \phi)}{g \cos^2 \phi} \right)$$

Path of projectile



$$\text{let } x = d \cos \phi$$

$$y_{\text{max}} = d \sin \phi$$

$$y_{\text{max}} = vt - \frac{1}{2}gt^2$$

$$d \sin \phi = vt - \frac{1}{2}gt^2$$

$$y_{\text{max}} = d \sin \phi$$

$$d \sin \phi = v \sin \theta_0 t - \frac{1}{2}gt^2$$

$$d \sin \phi =$$

$$d \sin \phi = \frac{v \sin(\theta_0 - \phi)}{v \cos \theta_0} - \frac{1}{2}g \left(\frac{d \cos \phi}{v \cos \theta_0} \right)^2$$

$$d \sin \phi = v$$

$$d \sin \phi = \frac{d g \cos \theta_0 \sin \theta_0 - \frac{1}{2} d^2 \cos^2 \theta_0}{v \cos \theta_0}$$

$$d \sin \phi = \frac{d (\cos \theta_0 \sin \theta_0 - \frac{1}{2} \cos^2 \theta_0)}{v \cos \theta_0}$$

$$\sin \phi = \frac{\cos \theta_0 \sin \theta_0 - \frac{1}{2} \cos^2 \theta_0}{v \cos \theta_0}$$

$$\sin \phi \cos \theta_0$$

Gr. Physics - I . Mid-exam

ADAMA SCIENCE AND TECHNOLOGY UNIVERSITY

DEPARTMENT OF PHYSICS

GENERAL PHYSICS-I MID -EXAM

NAME _____

ID.NO. _____

GROUP _____

TIME ALLOWED: 2hrs

Directions:

This exam booklet contains two parts: Part I is the short answer type questions [16%] and Part II is the workout questions [24%]. Attempt all the questions and write your answers legibly.

FOR INSTRUCTORS USE ONLY

Part I	Part II	Total

1/1

I: Give short answers for questions's below [16pts]

1) Given two vectors: $\vec{A} = 2\hat{i} - 3\hat{j}$ and $\vec{B} = -\hat{i} + 8\hat{k}$. Find:
a) the angle between the two vectors. [2pt]

b) The projection of $\vec{A}$ along $\vec{B}$ [2pt]

2) For two vectors: $\vec{A} = 4\hat{i} + 3\hat{j} - \hat{k}$ and $\vec{B} = 2\hat{i} - 3\hat{j} + 2\hat{k}$ obtain a unit vector normal to the plane formed by the two vectors. [2pt]

3) For one dimensional motion, is that always true that acceleration and velocity have the same direction? Explain. [2pt]

4) A ball is kicked with 20m/s at an angle of 53° with +x-axis.
a) What is its velocity at maximum height? [2pt]

b) What is its displacement at the end of 0.5 second? [2pt]

5) A block of mass 5kg is pulled by a force of 20N, making an angle of 37° with the horizontal axis. If the coefficient of friction between the block and the horizontal surface is 0.1, determine the magnitude of acceleration of the block. [2pt]

6) A rock and a ball which are at the same initial elevation are set into motion simultaneously in such a way that the rock is thrown horizontally while the ball is released freely. Which one reaches the ground first? justify [2pt]

1

2

II: Workout. Show your steps clearly [24pts]

1) Find the angle of a projection for which the horizontal range and the maximum height of the projectile are equal. [4pt]

2) A particle initially located at the origin has an acceleration of $\mathbf{a} = (3\hat{j}) \frac{m}{s^2}$ and an initial velocity of $\mathbf{v}_0 = (5\hat{i}) \frac{m}{s}$. Find:

a) the position vector at any time t [2pt]

b) velocity of the particle at $t = 2s$. [3pt]

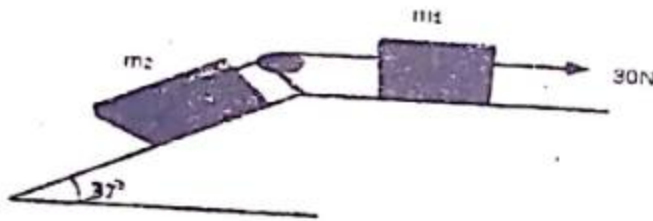
3) position vector of a particle is given as: $\vec{r} = (2t + 1)\hat{i} + t^2\hat{j}$, where r is in meter and t is in second. Find:

a) average velocity in the time intervals $t_1 = 2s$ and $t_2 = 4s$ [3pt]

b) The instantaneous acceleration at $t = 3s$ [3pt]

4) The World's Food Organization is delivering medicine and food to a small village by air-drop. If the plane is traveling at 6 m/s to the East at an altitude of 300 m , how far in advance of the village should the airplane need to be to drop the food so that the package will land in the village? [3pt]

5) Two blocks of masses $m_1 = 4\text{ kg}$, and $m_2 = 2\text{ kg}$ are pulled by 30 N force as shown in the figure below. If the inclined plane is frictionless, and if the pulley is light and frictionless, find:



a) The maximum value of the coefficient of static friction between m_1 and the horizontal surface that keeps the system at rest. [3pt]

b) The tension in the coupling [3pt]

Gen. Physics-I Exam

ADAMA SCIENCE AND TECHNOLOGY UNIVERSITY

SCHOOL OF NATURAL SCIENCES

DEPARTMENT OF PHYSICS

EXAM: GENERAL PHYSICS I (Phys101)

Time given 3: hrs.

NAME _____

Group _____

GENERAL INSTRUCTION: THIS EXAM BOOKLET CONTAINS TWO PARTS:

PART I- Short Answers. Check there are 11 short answer questions out of 25%.

PART II- There are 5 work out problems out of 30%.

Attention: Check Totally there're 9 Pages for this Exam.

ATTEMPT ALL QUESTIONS ACCORDINGLY. AND WRITE ALL NECESSARY STEPS FOR THE WORKOUT PART.

FOR INSTRUCTORS USE ONLY

Short Answer (1 to 11)	Total Mark 55%					
	1	2	3	4	5	

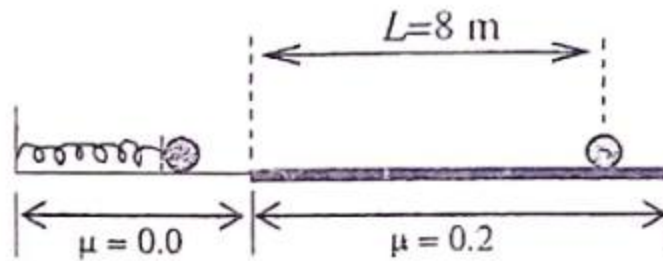
PART I: Short Answers (25 Pts)

1. If the speed of an object is halved, then its kinetic energy can be changed by the factor of _____ (2pt).
2. Write at least two properties of conservative forces (3pt)
 - a. _____
 - b. _____
3. Two cars of masses $m_1 = 3000\text{kg}$ and $m_2 = 2000\text{kg}$ were moving at velocities $v_1 = 20\text{m/s } \hat{i}$ and $v_2 = 15\hat{j} \text{ m/s}$, respectively. The velocity of their centers of mass is _____ (2pt)
4. A system of particles consists of 2 kg mass located at (1, 2, 3) m, 3 kg mass at (-2, 1, 4) m and 1 kg mass at (-1, 0, 8) m. The coordinate of the center of ~~mass~~ ^{mass} is _____ (2pts)
5. A constant force is acting on a particle that was moving along a space curve so that the angular momentum $\vec{L} = (2 \text{ kgm/s})\hat{i}$ remains a constant. What would be the value of torque acting on the bod? _____ (2pt)
6. An object of mass 3 kg moving with an initial velocity of $5\hat{i} \text{ m/s}$ collides with and sticks together to an object of mass 2 kg which initially move with velocity of $-3\hat{j} \text{ m/s}$. The final velocity of the composite object is _____ (2pts)
7. A force $\vec{F} = (6\hat{i} + 5\hat{j}) \text{ N}$ acts on an object as the object moves in x-direction from an origin to 5 m. The work done by the force is _____ (2pt)
8. What can be said about the speed of an object if the net work done on it is zero? (3pt)
9. State work energy theorem. (2 pt)
10. A uniform thin spherical shell ($I_{cm} = \frac{2}{3}MR^2$) and hoop ($I_{cm} = MR^2$) are placed side by side at the top of an inclined plane of height "h". If they are released from rest and without slipping, which one reaches the bottom first? Explain your answer. (3 pt)

11. A spring A and B are identical except that A is stiffer than B, i.e. $K_A > K_B$. In which spring is more work is expended, if (2pt)
- They are stretched by the same amount _____
 - They are stretched by the same amount of force _____

PART I: Workout problems. Show all necessary steps (Each 6 Pts)

- 1 A 3 kg-object is released from a compressed spring whose force constant is 120 N/m. After leaving the spring it travels over a horizontal surface, with coefficient of kinetic friction of 0.2, for a distance of 8m before coming to rest as shown in Figure below.



- (a) What was its maximum kinetic energy?

(b) How far was the spring compressed before being released?

- 2 A 100 g rubber ball initially at rest drops vertically from a height of 10 m on to a hard floor and it rebounds with a speed of v_0 . If the ball contacts with floor for 40 milliseconds and work done by the floor is 7 J;
- (a) What is the impulse acted on the ball during the contact?

(b) What is the average force exerted on the ball by the floor?

(c) How high does the ball bounce?



solution

Gr. physics - I Mid-exam

(I) Short answer. (P Pts)

1) Given $V_0 =$ Required
 g $t = ?$

$$h = V_0 \sin \theta t - \frac{1}{2} g t^2$$

but at the ground $h = 0$

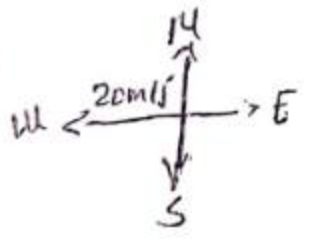
$$0 = V_0 \sin \theta t - \frac{1}{2} g t^2$$

$$\frac{1}{2} g t^2 = V_0 \sin \theta t$$

$$g t^2 = 2 V_0 \sin \theta t$$

$$g t = 2 V_0 \sin \theta$$

$$t = \frac{2 V_0 \sin \theta}{g}$$

2)  The resultant force is 0
B/c as a velocity constant
acceleration is zero so
 $F = ma \Rightarrow \underline{F = 0}$ b/c $V_f = V_i$

(3) $p = mv$ $V_2 = 2V_1$ so if speed doubled
momentum also double

$$p_1 = m 2V_1$$

$$p_2 = 2mV_1$$

$$KE = \frac{1}{2} m V^2$$

$$K_2 = \frac{1}{2} m (2V_1)^2$$

$$\frac{-1-}{14}$$

$$2KE_1 = mV^2$$

$$V^2 = \frac{2KE_1}{m}$$

$$2KE_2 = 4mV^2$$

$$V^2 = \frac{2KE_2}{4m}$$

①.1

$$\Rightarrow V^2 = V^2$$

$$\frac{2KE_1}{m} = \frac{2KE_2}{4m}$$

$$\underline{KE_2 = 4KE_1}$$

increased by 4

(4) state and explain?

(a) work energy theorems

$\Rightarrow$ states that the net work done on an object by the net force is equal to the change in kinetic energy of the object

$$W_{\text{net}} = E_{Kf} - E_{Ki}$$

(b) newton's law of motion

1st law states that - every particle ~~persists~~ in a state of rest or uniform motion in straight line (with constant velocity) unless acted upon by unbalanced force.

* This law of motion is the law of "inertia"

2nd law states that $\Rightarrow$ If the external force $\vec{F}$ is acting on a particle of mass m which as a consequence is moving with a velocity $\vec{V}$ then
$$\vec{F} = \frac{d}{dt}(m\vec{V}) = m \frac{d\vec{V}}{dt} = m\vec{a}$$

3rd law states that $\Rightarrow$ when two forces are act on a body the force is equal in magnitude but opposite in direction.
"action and reaction" no movement

(c) properties of conservative force $\Rightarrow$

- 1 - A force is conservative if the work it does on a particle moving b/n any two points is independent of the path taken by particle
- 2 - The work done by conservative force on a particle moving through any closed path is zero

(d) center of mass of the system of particles $\Rightarrow$

$\Rightarrow$ The linear momentum of the system of particle is equal to the product of the total mass M of the system and the velocity of the center of mass

- This expression shows that if the net external force acting on the system of particles is zero ($F_{\text{ext}} = 0$)
- 2 - Linear momentum is conserved

II. WORKOUT PROBLEMS

1) $V = 3t\hat{i} - 4t^2\hat{j} + 2\hat{k}$

(a) displacement at $t=1$
 $t_2=4$

$$\vec{r}_2 - \vec{r}_1 = \vec{D}$$

$$r(t_2) - r(t_1) = \vec{D}$$

$$3(4) - 4(4)^2 + 2 - (3(1) - 4(1)^2 + 2)$$

$$12 - 4(16) + 2 - (3 - 4 + 2)$$

$$14 - 64 - (1)$$

$$-51 = \underline{\underline{-51 \text{ unit to } -x\text{-axis}}}$$

$$(c) a_x = \frac{V_t - V_0}{t_2 - t_1} = \frac{3 - 8t_2 - (3 - 8t_1)}{t_2 - t_1}$$

$$= \frac{-21 - (-5)}{2} = \frac{-21 + 5}{2}$$

$$= \frac{-16}{2} = \underline{\underline{-8 \text{ unit per second square}}}$$

(d) a_{inst} at time t

$$\frac{d^2 r}{dt^2} = 3t\hat{i} - 4t^2\hat{j} + 2\hat{k}$$

$$= 3 - 8t$$

$$= \underline{\underline{-8 \text{ unit per second square}}}$$

$$(b) \frac{dr}{dt} = 3t\hat{i} - 4t^2\hat{j} + 2\hat{k}$$

$$V = 3 - 8t$$

$$= 3 - 8(3)$$

$$= 3 - 24 = \underline{\underline{-21 \text{ unit per second}}}$$

$$\tan \theta = \frac{y}{x}$$

$$\theta = \tan^{-1}\left(\frac{-21}{3}\right)$$

$$= \tan^{-1}(-7)$$

$$= 360^\circ - \tan^{-1}(7)$$

$$= 360^\circ - 81.87^\circ$$

$$= \underline{\underline{278.13^\circ}}$$

2) $d = \frac{\sqrt{2}V_0^2 \cos \theta_0 \sin(\theta_0 - \phi)}{g \cos^2 \phi}$

I - Give short answer (16 pts)

1) Given $A = 2\hat{i} - 3\hat{j}$ Find $\theta - \theta$
 $B = -\hat{i} + 8\hat{k}$

b - Projection A along B
(mean $A \times B$)

$$a) \cos \theta = \frac{A \cdot B}{|A||B|}$$

$$\cos \theta = \frac{(2\hat{i} - 3\hat{j} + 0\hat{k}) \cdot (-\hat{i} + 0\hat{j} + 8\hat{k})}{\sqrt{4+9} \cdot \sqrt{1+64}}$$

$$= \frac{-2}{\sqrt{13} \cdot \sqrt{65}} = \frac{-2}{227.2} = -0.0088$$

$$\theta = \cos^{-1}(0.0088) \\ = \underline{\underline{89.5^\circ}}$$

$$A \times B = \begin{bmatrix} 0 & 0 & 16 \\ 2 & -3 & 0 \\ -1 & 0 & 8 \end{bmatrix}$$

$$= \underline{\underline{-24\hat{i} - 16\hat{j} - 3\hat{k}}}$$

2) Normal unit Vector $= \frac{A \times B}{|A||B|} = \frac{A \times B}{|A \times B|}$

$$A \times B = \begin{bmatrix} 0 & 0 & 16 \\ 2 & -3 & 0 \\ -1 & 0 & 8 \end{bmatrix}$$

$$A \times B = 3\hat{i} - 10\hat{j} - 18\hat{k}$$

$$\text{Normal (unit)} = \frac{3\hat{i} - 10\hat{j} - 18\hat{k}}{\sqrt{9+100+324}} \\ = \frac{3\hat{i} - 10\hat{j} - 18\hat{k}}{\sqrt{433}}$$

$$= \underline{\underline{\frac{1}{20.8} (3\hat{i} - 10\hat{j} - 18\hat{k})}}$$